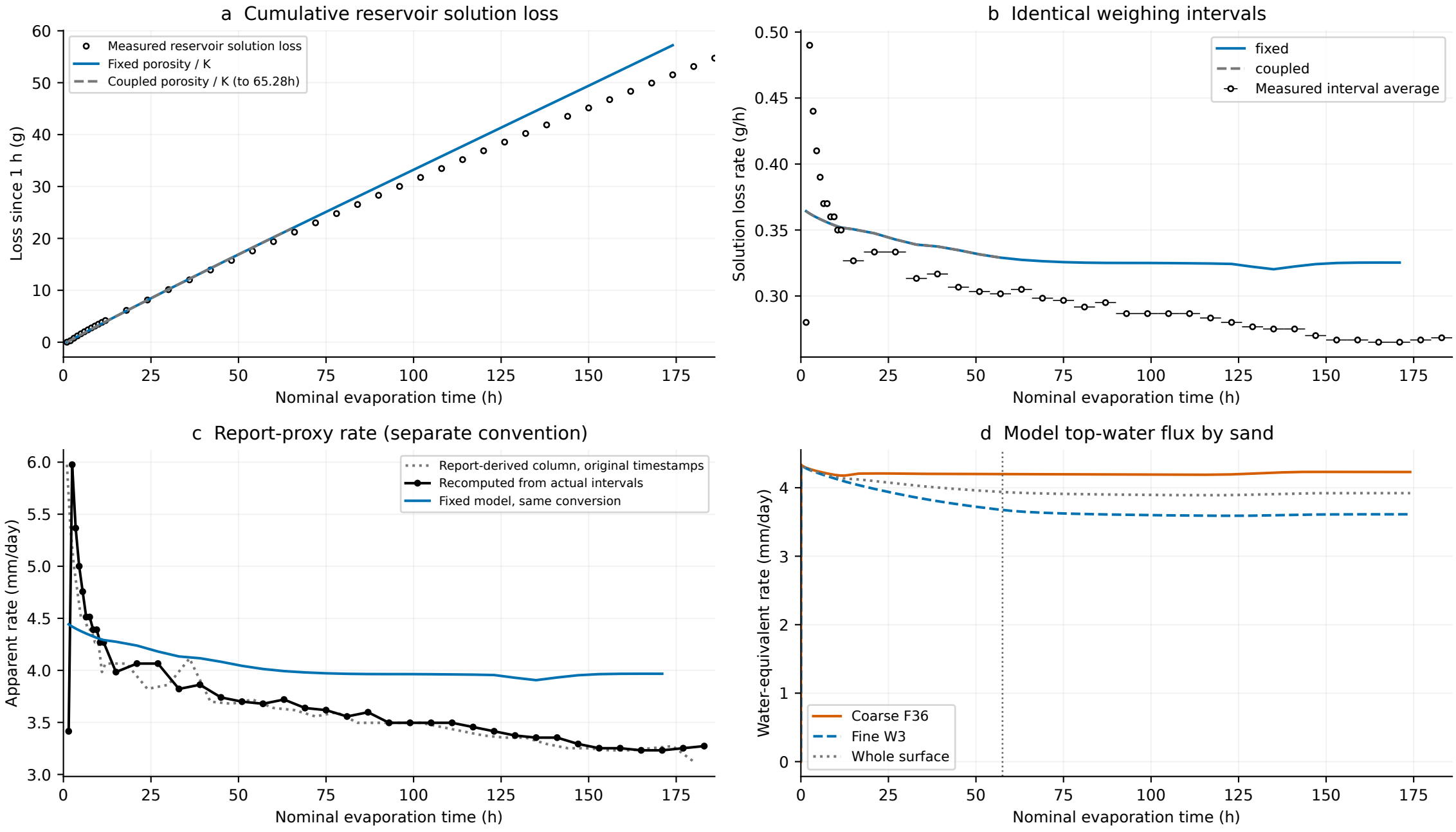
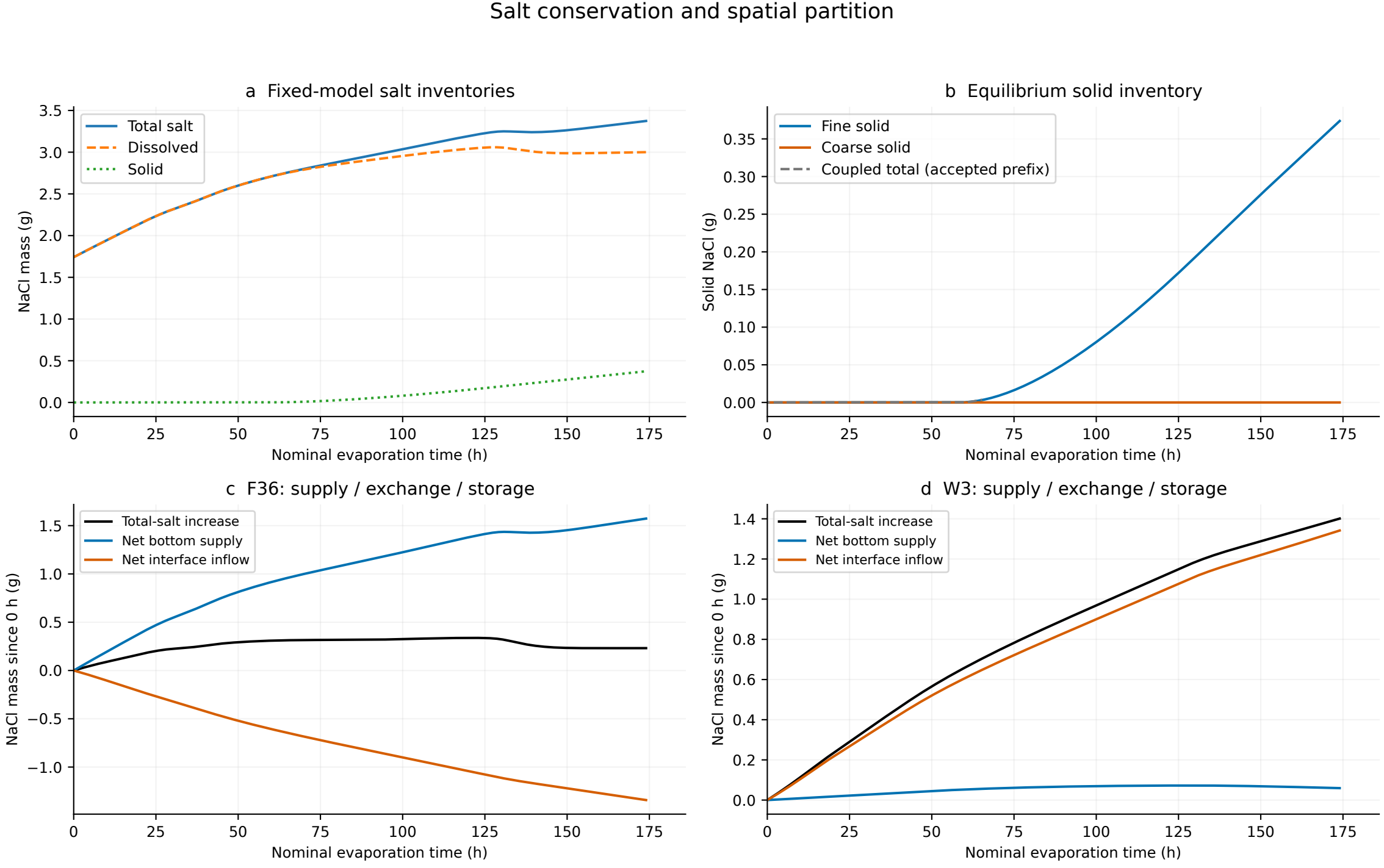
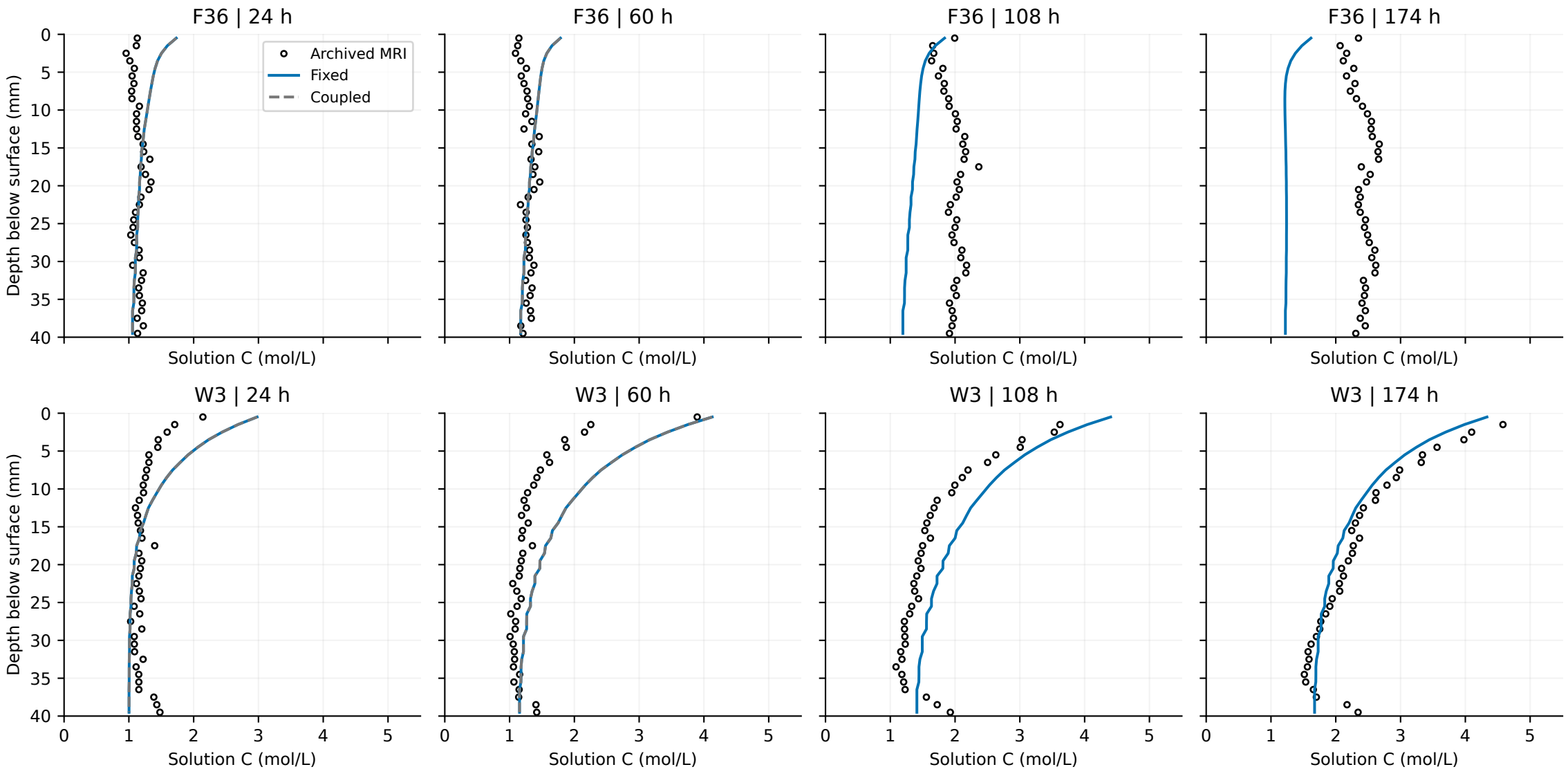
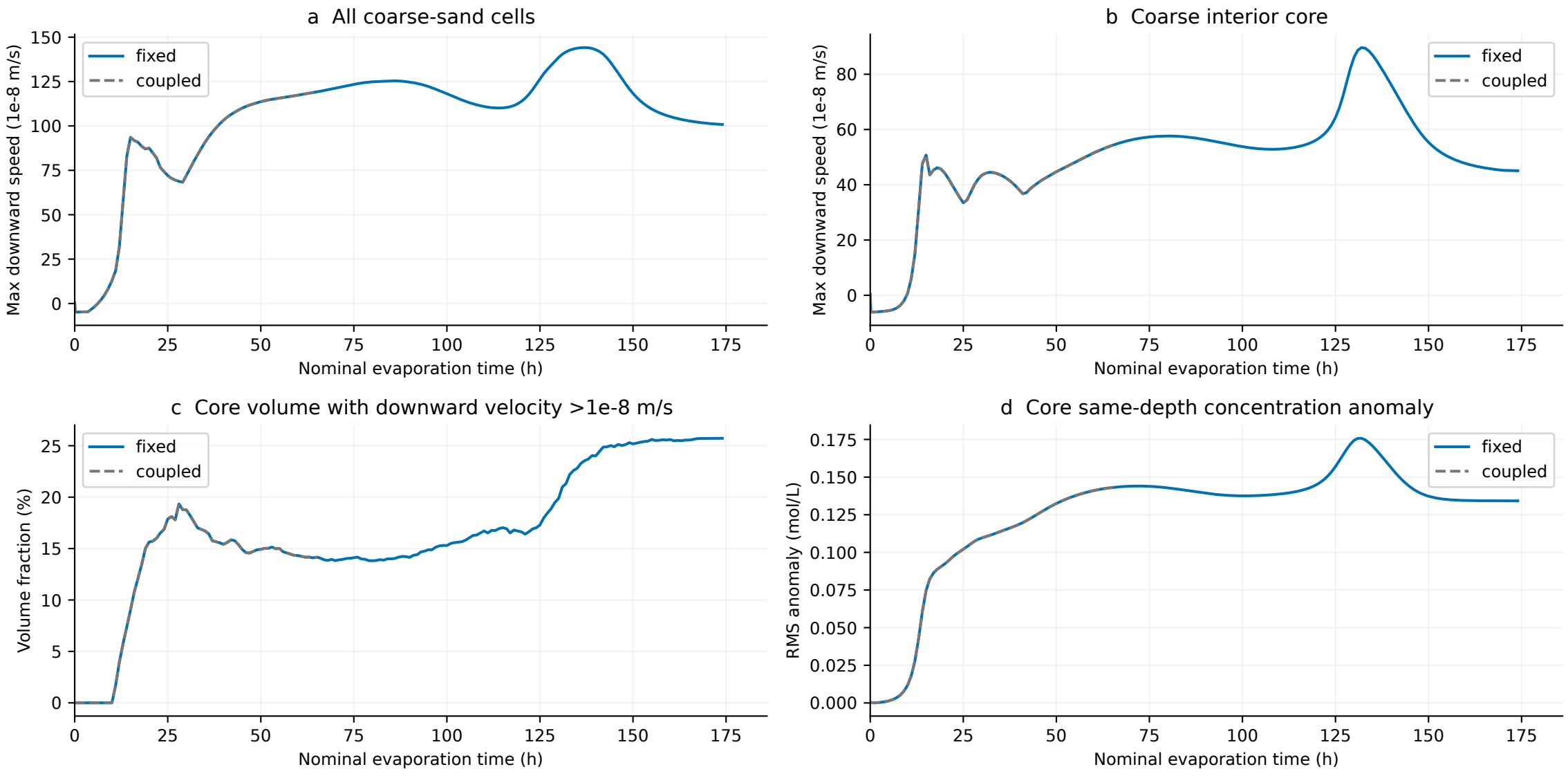


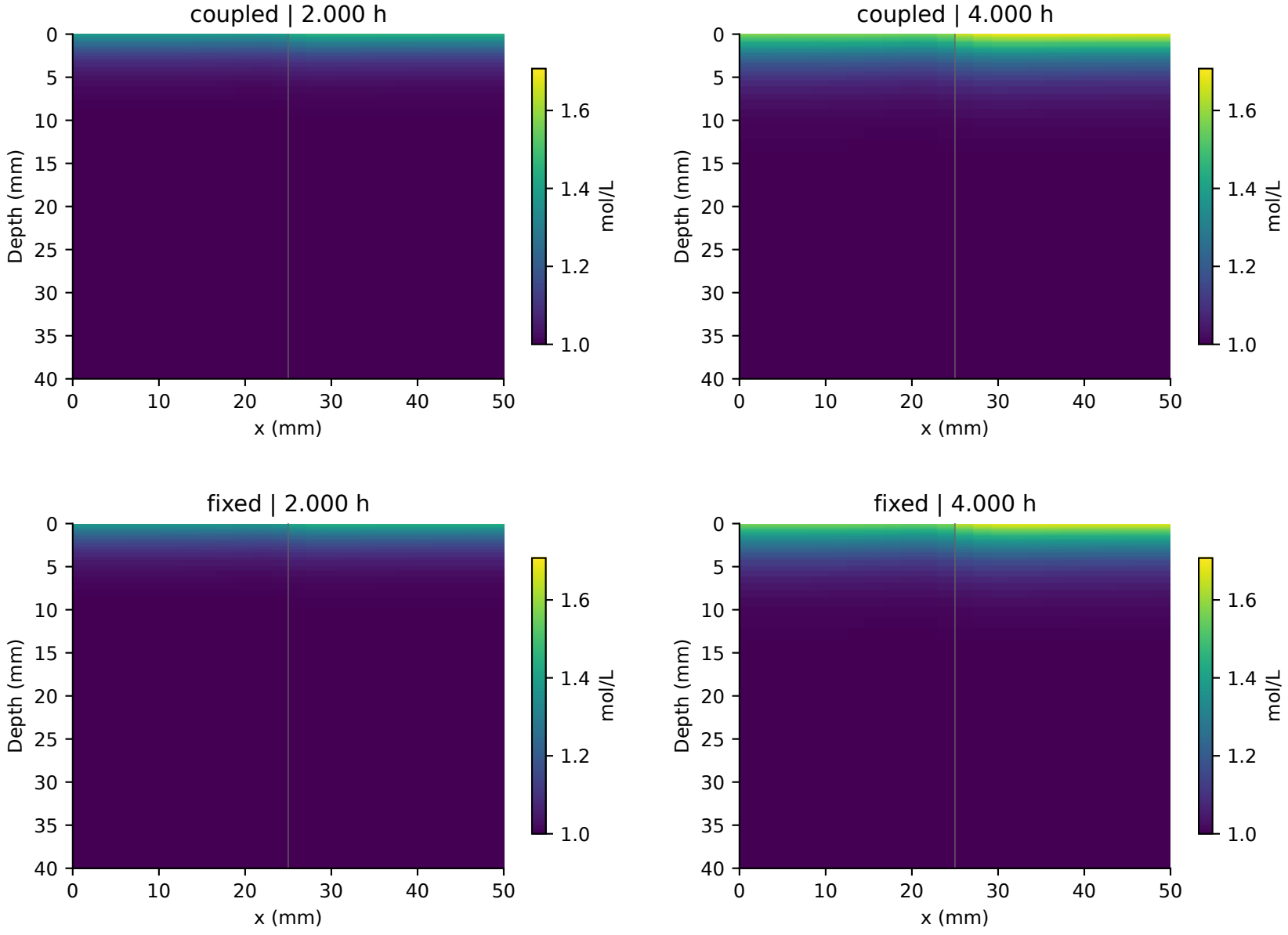
No extrapolation of the failed coupled run. Horizontal bars are measurement intervals, not uncertainty; no measured t=0 mass.

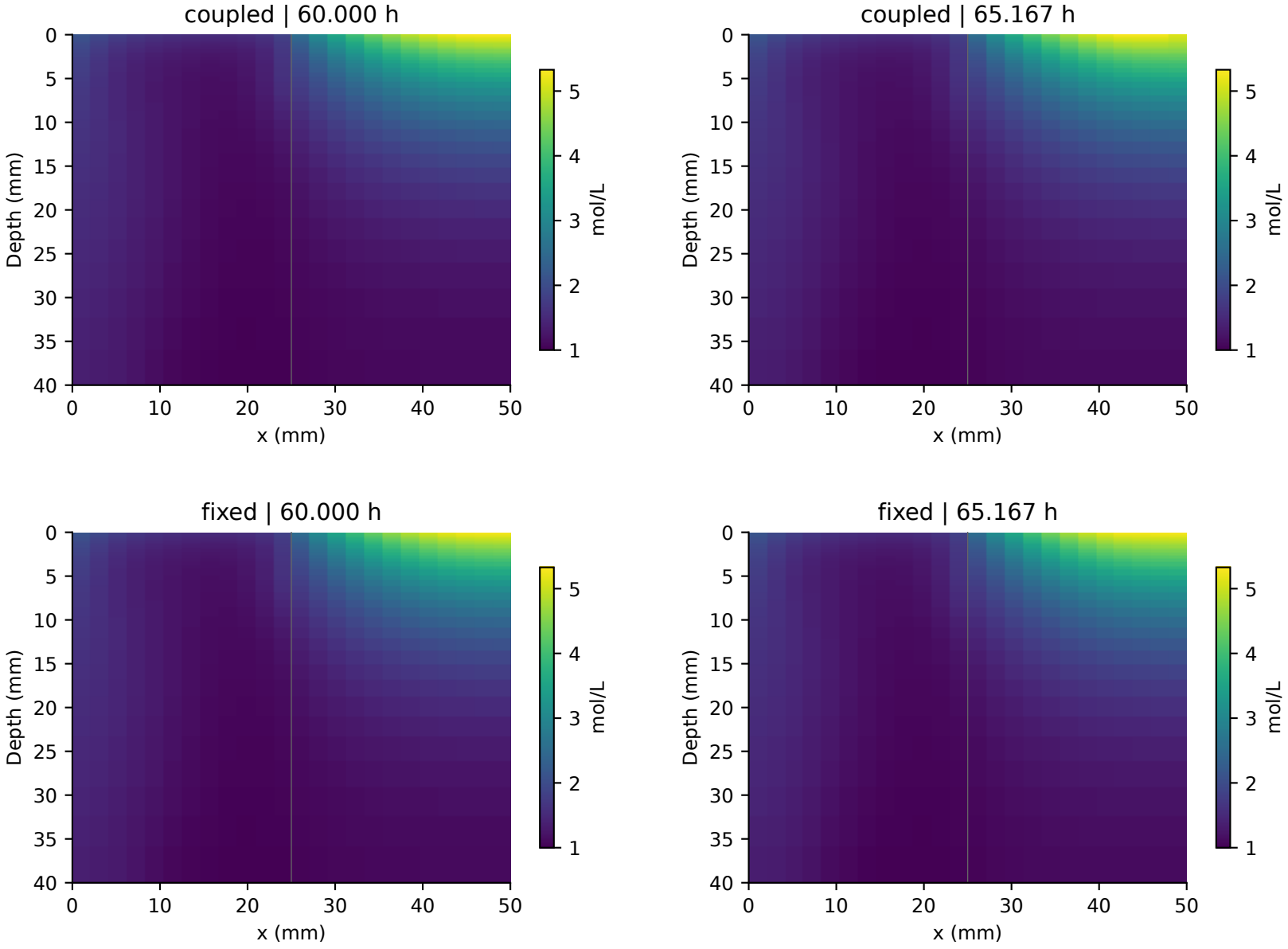

Salt conservation and spatial partition
a Fixed-model salt inventories
Total salt
Dissolved
Solid
NaCl mass (g)
Nominal evaporation time (h)
b Equilibrium solid inventory
Fine solid
Coarse solid
Coupled total (accepted prefix)
Solid NaCl (g)
Nominal evaporation time (h)
c F36: supply / exchange / storage
Total-salt increase
Net bottom supply
Net interface inflow
NaCl mass since 0 h (g)
Nominal evaporation time (h)
d W3: supply / exchange / storage
Total-salt increase
Net bottom supply
Net interface inflow
NaCl mass since 0 h (g)
Nominal evaporation time (h)
Solid is retained in total-salt storage. Equivalent solid volume is an inventory diagnostic, not geometric pore occupancy or salt-crust shape.

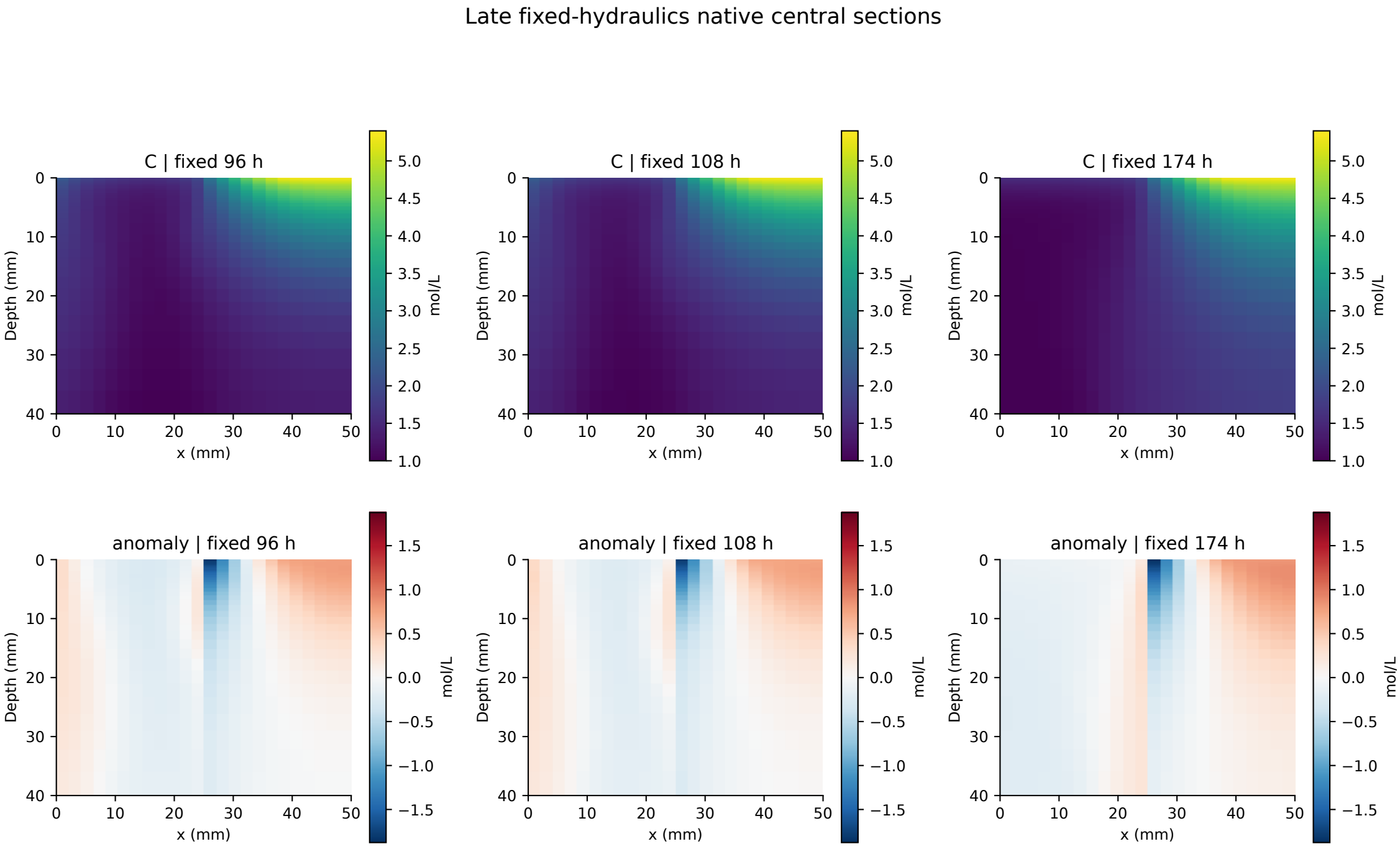


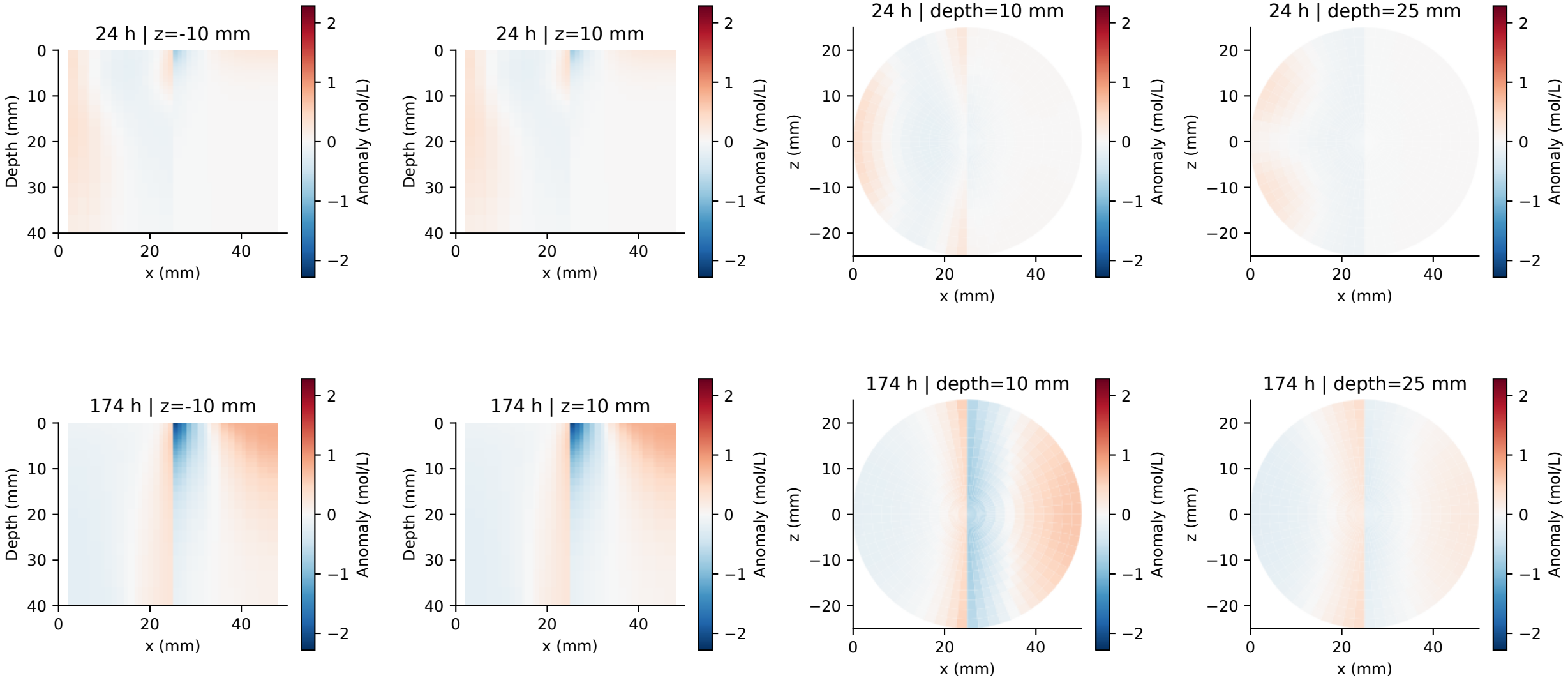
MRI is dissolved-salt evidence, not total salt or crystallization timing. Exact voxel processing / 32 min scan window are not reproduced; nominal clock.


Coarse-sand downward-flow diagnostics
a All coarse-sand cells
b Coarse interior core
c Core volume with downward velocity >1e-8 m/s
d Core same-depth concentration anomaly
fixed
coupled
Max downward speed (1e-8 m/s)
Volume fraction (%)
RMS anomaly (mol/L)
Nominal evaporation time (h)
Core excludes wall/top/bottom 5 mm and interface 3 mm. Interior-LS velocity; hourly after 4 h; threshold is not MRI-visible or linear-instability onset.


Early central sections: unchanged pre-precipitation prefix
coupled | 2.000 h
coupled | 4.000 h
fixed | 2.000 h
fixed | 4.000 h
Depth (mm)
x (mm)
mol/L
0
5
10
15
20
25
30
35
40
0
10
20
30
40
50
1.0
1.2
1.4
1.6
Shared color scale; native finite-volume cells at z=0. Coarse sand left, fine sand right. No coupled-model fields beyond its saved prefix.


Last common native-field comparison after equilibrium precipitation
coupled | 60.000 h
coupled | 65.167 h
fixed | 60.000 h
fixed | 65.167 h
Depth (mm)
x (mm)
mol/L
0
5
10
15
20
25
30
35
40
0
10
20
30
40
50
1
2
3
4
5
Shared color scale; native finite-volume cells at z=0. Coarse sand left, fine sand right. No coupled-model fields beyond its saved prefix.


Late fixed-hydraulics native central sections
C | fixed 96 h
C | fixed 108 h
C | fixed 174 h
anomaly | fixed 96 h
anomaly | fixed 108 h
anomaly | fixed 174 h
Depth (mm)
x (mm)
mol/L
Same-depth anomaly removes each material layer mean. Late fields have no completed coupled counterpart and do not establish experimental plume agreement.


Off-centre and horizontal fixed-model concentration structure
24 h | z=-10 mm
24 h | z=10 mm
24 h | depth=10 mm
24 h | depth=25 mm
174 h | z=-10 mm
174 h | z=10 mm
174 h | depth=10 mm
174 h | depth=25 mm
Depth (mm)
z (mm)
x (mm)
Anomaly (mol/L)
Common anomaly scale at 24 and 174 h; whole-half MRI profiles cannot validate these individual cross-sections.